FILTER DESIGN TEST CASE

Band-Pass Filter - n77 (3.3-4.2 GHz 5G NR) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Band-Pass Filter filter targeting n77 (3.3-4.2 GHz 5G NR) for Automotive V2X Communication. The filter is designed on AlN on SiC substrate with a target center frequency of 1054.3 MHz and bandwidth of 227.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Band-Pass Filter
- Frequency Band: n77 (3.3-4.2 GHz 5G NR)
- Application: Automotive V2X Communication
- Substrate: AlN on SiC
- Package: CSP (Chip Scale Package)
- Design Tool: COMSOL Multiphysics 6.2
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Band-Pass Filter design targeting n77 (3.3-4.2 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 1054.3 MHz
Bandwidth (3 dB): 227.4 MHz
Insertion Loss Target: <= 1.1 dB
Return Loss Target: >= 19.0 dB
Out-of-Band Rejection: >= 38.0 dB
Isolation (Duplexer): >= 34.0 dB
Q Factor: 7694
Temperature Coefficient:-21.7 ppm/°C
Die Size: 1.4 x 2.3 mm
Wafer Size: 8-inch
Number of Resonators: 7
Electrode Material: Mo/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open COMSOL Multiphysics 6.2 and load the Band-Pass Filter template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 1054.3 MHz, BW 227.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.1 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on SiC).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 3163 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.1 dB
- Return loss in passband shall be >= 19.0 dB
- Out-of-band rejection at +/- 341 MHz offset >= 38.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 2.9 MHz
- Power handling shall be >= +25 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 86%

6. TEST RESULTS

Measurement Date: 2024-04-16
Devices Tested: 42
Insertion Loss (meas): 0.84 dB
Return Loss (meas): 21.4 dB
Rejection (meas): 41.4 dB
Isolation (meas): 39.5 dB
Q Factor (meas): 7694
Group Delay Variation: 5.6 ns
Power Handling: +30 dBm
Die Yield: 97.8%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

a) The Band-Pass Filter design demonstrated below-target performance for n77 (3.3-4.2 GHz 5G NR).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Temperature stability was within specification across full range.
e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

Author: Dr. Mei-Ling Chen Date: 2024-04-16
Reviewer: Dr. Hiroshi Tanaka Date: 2024-04-19
Approver: Dr. James Liu Date: 2024-04-25

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